

Data Extraction

EXPLORER

SOURCE CONTROL: REPOSITORIES

project

SOURCE CONTROL: CHANGES

PROJECT

- > __pycache__
- > .dvc
- > airflow
- > dags
- > data
 - > processed
 - > raw
 - .gitkeep
- stock_data_AAPL_20251201_164801.csv
- models
- reports
- scripts
 - data_quality_check.py

OUTLINE

TIMELINE

SOURCE CON... Auto

The selected source control provider does not have any source control history items.

data > raw > stock_data_AAPL_20251201_164801.csv

```
1 ,open,
2 2025-11-28 08:40:00,278.05,278.49,277.01,277.25,69693,AAPL,2025-11-28
3 2025-11-28 08:45:00,278.3012,278.64,278.3012,278.59,3205,AAPL,2025-11-28
4 2025-11-28 08:50:00,278.55,278.6,277.55,278.53,240597,AAPL,2025-11-28
5 2025-11-28 08:55:00,278.51,278.6,278.3,278.43,5366,AAPL,2025-11-28
6 2025-11-28 09:00:00,278.43,278.6,278.3,278.537,1340,AAPL,2025-11-28
7 2025-11-28 09:05:00,278.46,278.59,278.3072,278.53,2297,AAPL,2025-11-28
8 2025-11-28 09:10:00,278.5736,278.6,278.45,278.49,5443,AAPL,2025-11-28
9 2025-11-28 09:15:00,278.52,278.59,278.3,278.4,4559,AAPL,2025-11-28
10 2025-11-28 09:20:00,278.4137,278.49,278.0,278.0495,10017,AAPL,2025-11-28
11 2025-11-28 09:25:00,278.05,278.49,277.01,277.25,69693,AAPL,2025-11-28
```

PROBLEMS OUTPUT TERMINAL 24

JavaSE-21 LTS

```
(venv) D:\sem9\mlops\project>python scripts/extract_data.py
Fetching data for AAPL...
Using free tier: Last 100 data points (5-min intervals)
API Response Keys: ['Meta Data', 'Time Series (5min)']
Data extracted successfully: 100 records
Saved to: data/raw/stock_data_AAPL_20251201_164801.csv
Date range: 2025-11-28 08:40:00 to 2025-11-28 16:55:00

(venv) D:\sem9\mlops\project>python scripts/data_quality_check.py
data/raw/stock_data_AAPL_20241201_123456.csv

=====
Running Data Quality Checks
=====
File: data/raw/stock_data_AAPL_20241201_123456.csv
```

Data Wuality Checks Passed

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The selected source control provider does not have any source control history items.

data > raw > stock_data_AAPL_20251201_164801.csv

```
1 ,open,
2 2025-11-28 08:40:00,278.05,278.49,277.01,277.25,69693,AAPL,2025-11-28
3 2025-11-28 08:45:00,278.3012,278.64,278.3012,278.59,3205,AAPL,2025-11-28
4 2025-11-28 08:50:00,278.55,278.6,277.55,278.53,240597,AAPL,2025-11-28
5 2025-11-28 08:55:00,278.51,278.6,278.3,278.43,5366,AAPL,2025-11-28
6 2025-11-28 09:00:00,278.43,278.6,278.3,278.537,1340,AAPL,2025-11-28
7 2025-11-28 09:05:00,278.46,278.59,278.3072,278.53,2297,AAPL,2025-11-28
8 2025-11-28 09:10:00,278.5736,278.6,278.45,278.49,5443,AAPL,2025-11-28
9 2025-11-28 09:15:00,278.52,278.59,278.3,278.4,4559,AAPL,2025-11-28
10 2025-11-28 09:20:00,278.4137,278.49,278.0,278.0495,10017,AAPL,2025-11-28
11 2025-11-28 09:25:00,278.05,278.49,277.01,277.25,69693,AAPL,2025-11-28
```

PROBLEMS OUTPUT TERMINAL 24

JavaSE-21 LTS

```
(venv) D:\sem9\mlops\project>python scripts/data_quality_check.py
data/raw/stock_data_AAPL_20251201_164801.csv

=====
Running Data Quality Checks
=====
File: data/raw/stock_data_AAPL_20251201_164801.csv

✓ Checking minimum records (threshold: 50)...
✓ PASS: 100 records found

✓ Checking null values (threshold: 1.0%)...
✓ PASS: 'open' has 0.00% nulls
✓ PASS: 'high' has 0.00% nulls
✓ PASS: 'low' has 0.00% nulls
✓ PASS: 'close' has 0.00% nulls
✓ PASS: 'volume' has 0.00% nulls
```

Chat

Edit in context

AI responses may be inaccurate.

[Generate Agent Instructions](#) to onboard AI onto your codebase.

stock_data_AAPL_20251201_164801.csv

Edit or refactor selected code

Edit GPT-5 mini

Transforming Data into features and making processed data csv

The screenshot shows a JupyterLab environment. On the left, the 'SOURCE CONTROL: REPOSITORIES' panel is open, showing a project structure with folders like 'data', 'models', 'reports', and 'scripts'. The 'data' folder is expanded, showing files like 'processed_data_20251201_165514.csv' and 'stock_data_AAPL_20251201_164801.csv'. The central code editor displays a CSV file named 'stock_data_AAPL_20251201_164801.csv' with columns for date, time, and stock price. The terminal at the bottom shows the output of a data profiling report, indicating that the report has been generated and saved to 'reports/data_profile_20251201_165514.html'.

Stock Data Report

The screenshot shows a web browser displaying a 'Stock Data Profiling Report'. The report is titled 'Stock Data Profiling Report' and has tabs for 'Overview', 'Alerts', and 'Reproduction'. The 'Overview' tab is selected, showing a summary of the dataset statistics and variable types.

Dataset statistics	
Number of variables	40
Number of observations	80
Missing cells	0
Missing cells (%)	0.0%
Total size in memory	24.7 KiB
Average record size in memory	316.0 B

Variable types	
Numeric	38
Text	2

Below the tables, there is a 'Variables' section with a 'Select Columns' dropdown menu. The dropdown menu is currently set to 'open'.

Select Columns

Distinct	66	Minimum	276.21
Distinct (%)	82.5%	Maximum	278.86
Missing	0	Zeros	0
Missing (%)	0.0%	Zeros (%)	0.0%
Infinite	0	Negative	0
Infinite (%)	0.0%	Negative (%)	0.0%
Mean	277.7192563	Memory size	1.2 KiB



The screenshot displays the PyCharm IDE interface with three main panels:

- Project Tool Window (Left):** Shows the project structure. The `data` directory is expanded, showing `processed` (a folder) and `.gitkeep` (a file). The `processed` folder is highlighted, showing its contents: `processed_data_20251201_165514.csv` and `processed_data_20251201_165514.csv` (a file).
- Terminal Window (Bottom):** Displays the output of the `dvc push` command. The output shows the successful installation and uninstallation of `s3transfer` and `boto3`, followed by the successful installation of `aiobotocore`, `aiointertools`, `botocore`, `dvc-s3`, `s3fs`, and `wrapt`. The final output is `1 file pushed`.
- Source Control Tool Window (Bottom):** Shows the status of the source control system. The `initialized` button is visible, indicating that the source control system is initialized.

Model Registered in MLFLOW

The screenshot shows the MLflow web interface at localhost:5000. The experiment name is 'stock-price-prediction' under the 'Machine learning' category. The search filter is 'metrics.rmse < 1 and params.model = "tree"'. The table displays one run:

Run Name	Created	Dataset	Duration	Source	Models
stock_prediction_2025120...	46 minutes ago	-	16.0s	train.py	model

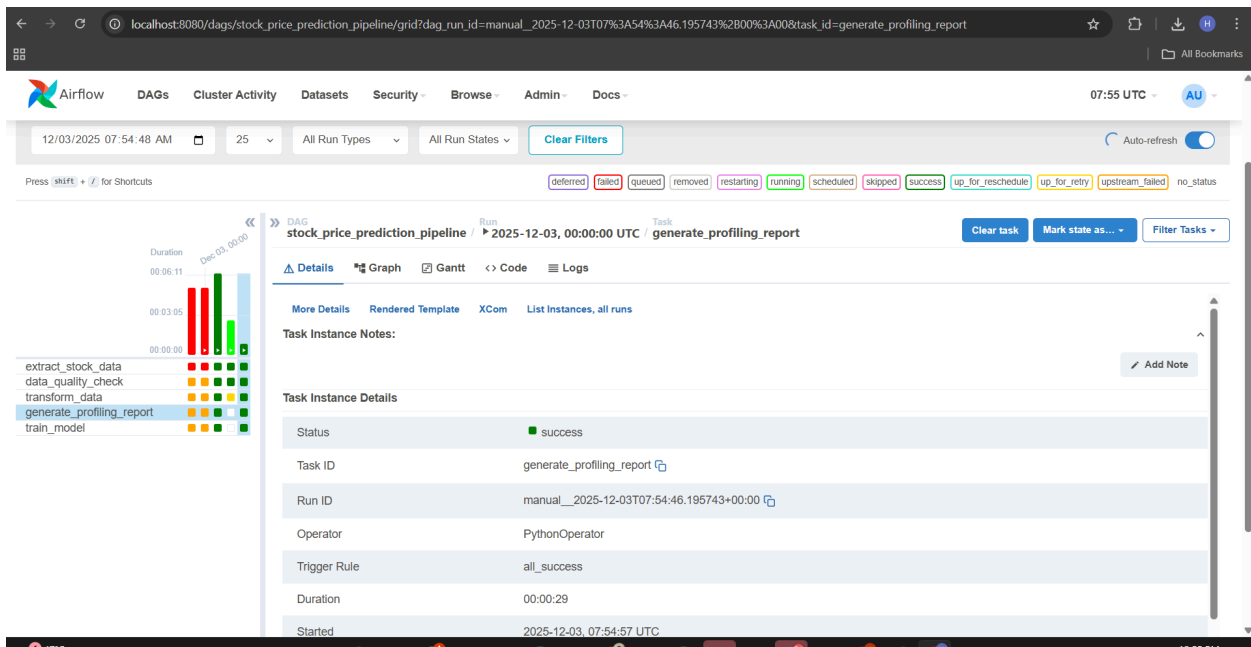
1 matching run

Experiments

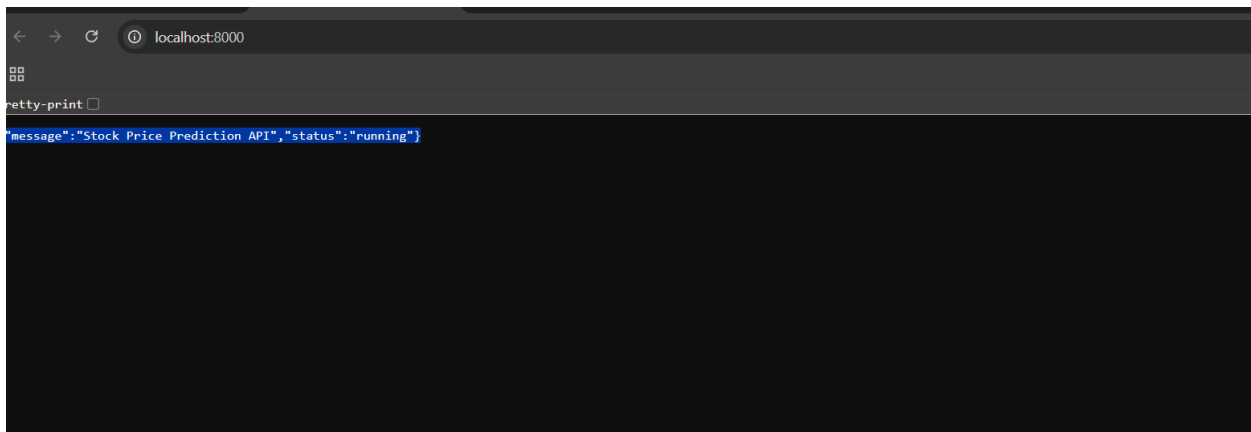
The screenshot shows the MLflow web interface at localhost:5001. The experiment name is 'stock-price-prediction'. The search filter is 'metrics.rmse < 1 and params.model = "tree"'. The table displays four runs:

Run Name	Created	Dataset	Duration	Source	Models
stock_prediction_2025120...	22 seconds ago	-	13.6s	train.py	sklearn
stock_prediction_2025120...	3 minutes ago	-	2.2s	train.py	-
stock_prediction_2025120...	4 minutes ago	-	1.7s	train.py	-
stock_prediction_2025120...	5 minutes ago	-	3.4s	train.py	-

Airflow dags running



Fastapi endpoint running



Model Loaded

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL 164 PORTS JavaSE-21 LTS
(venv) D:\sem9\mlops\project>docker logs mlops-fastapi
/usr/local/lib/python3.11/site-packages/pydantic/_internal/_fields.py:149: UserWarning: Field "model_version" has co
nflict with protected namespace "model_".

You may be able to resolve this warning by setting `model_config['protected_namespaces'] = ()`.
  warnings.warn(
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
⚠ Warning: Could not load model: [Errno 2] No such file or directory: '/app/models/model_latest.joblib'
INFO: 172.20.0.1:45002 - "GET / HTTP/1.1" 200 OK
INFO: 172.20.0.1:45002 - "GET /favicon.ico HTTP/1.1" 404 Not Found
INFO: 172.20.0.1:45002 - "GET / HTTP/1.1" 200 OK
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [1]
/usr/local/lib/python3.11/site-packages/pydantic/_internal/_fields.py:149: UserWarning: Field "model_version" has co
nflict with protected namespace "model_".

You may be able to resolve this warning by setting `model_config['protected_namespaces'] = ()`.
  warnings.warn(
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
✅ Model loaded from /app/models/model_latest.joblib
```

Docs

localhost:8000/docs#/default/predict_predict_post

"string"

GET /health Health

POST /predict Predict

Make prediction

Parameters

No parameters

Try it out

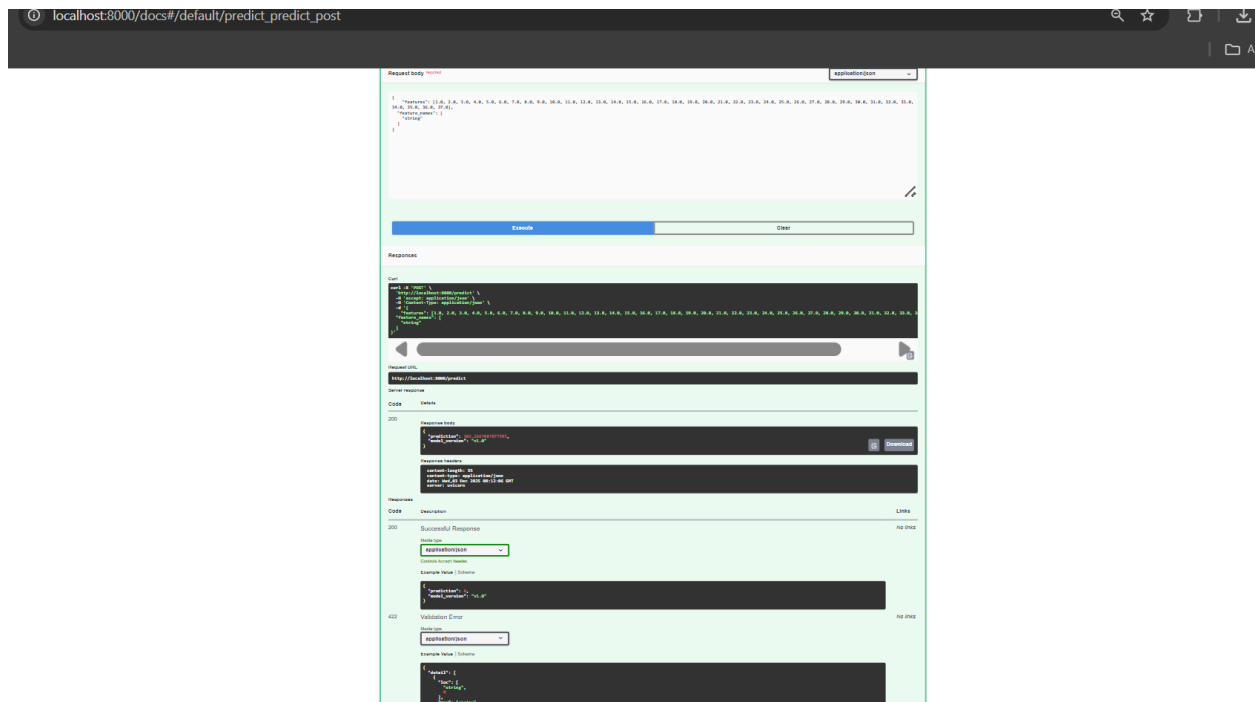
Request body required

application/json

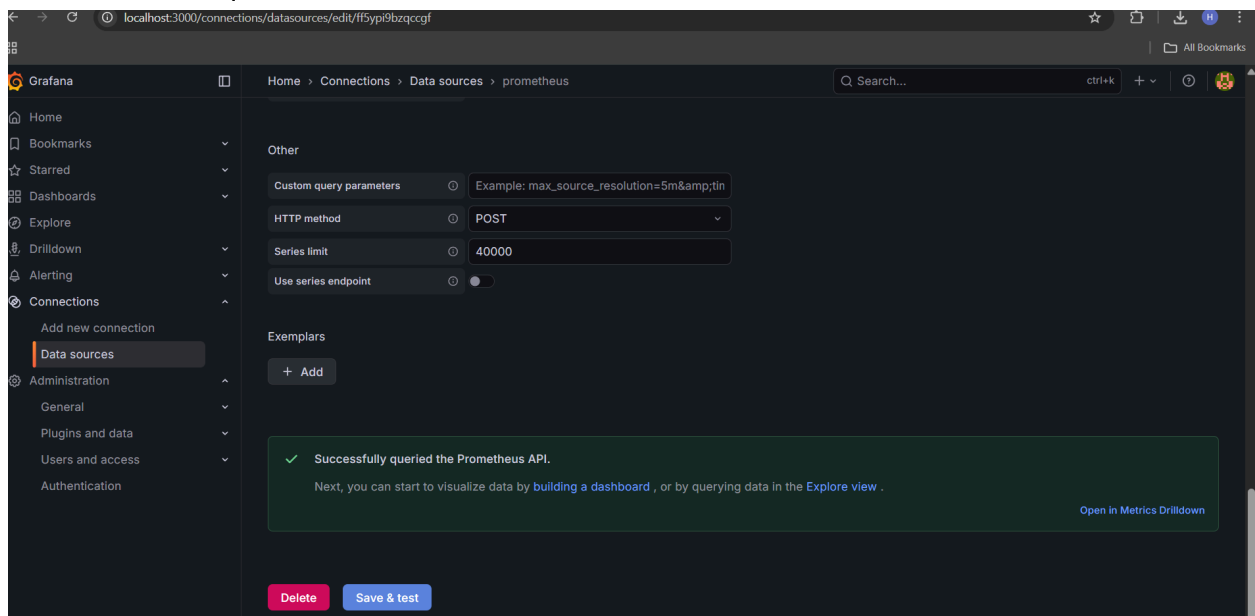
Example Value | Schema

```
{
  "features": [
    0
  ],
  "feature_names": [
    "string"
  ]
}
```

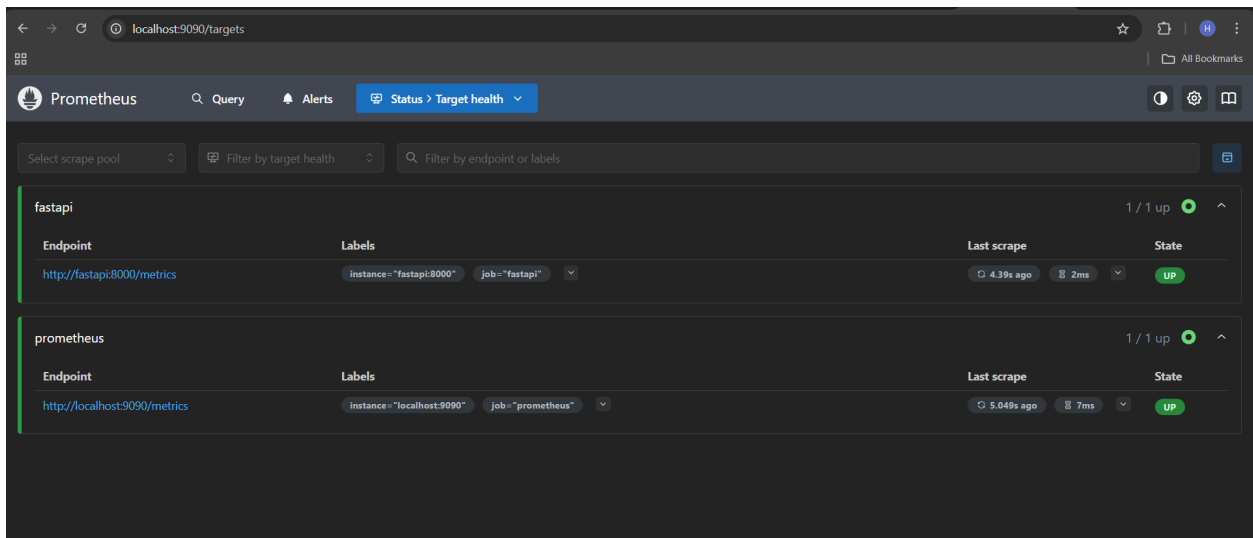
Prediction



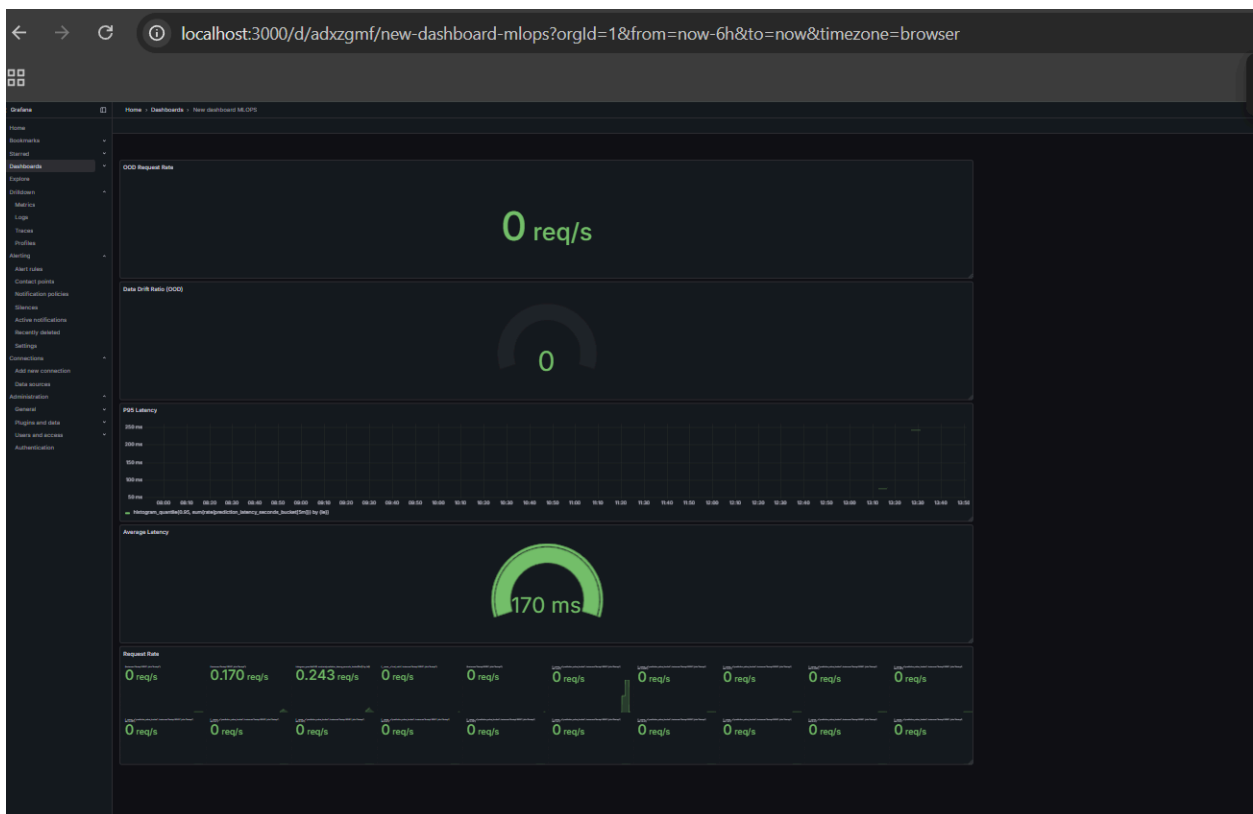
Prometheus Aoi queried



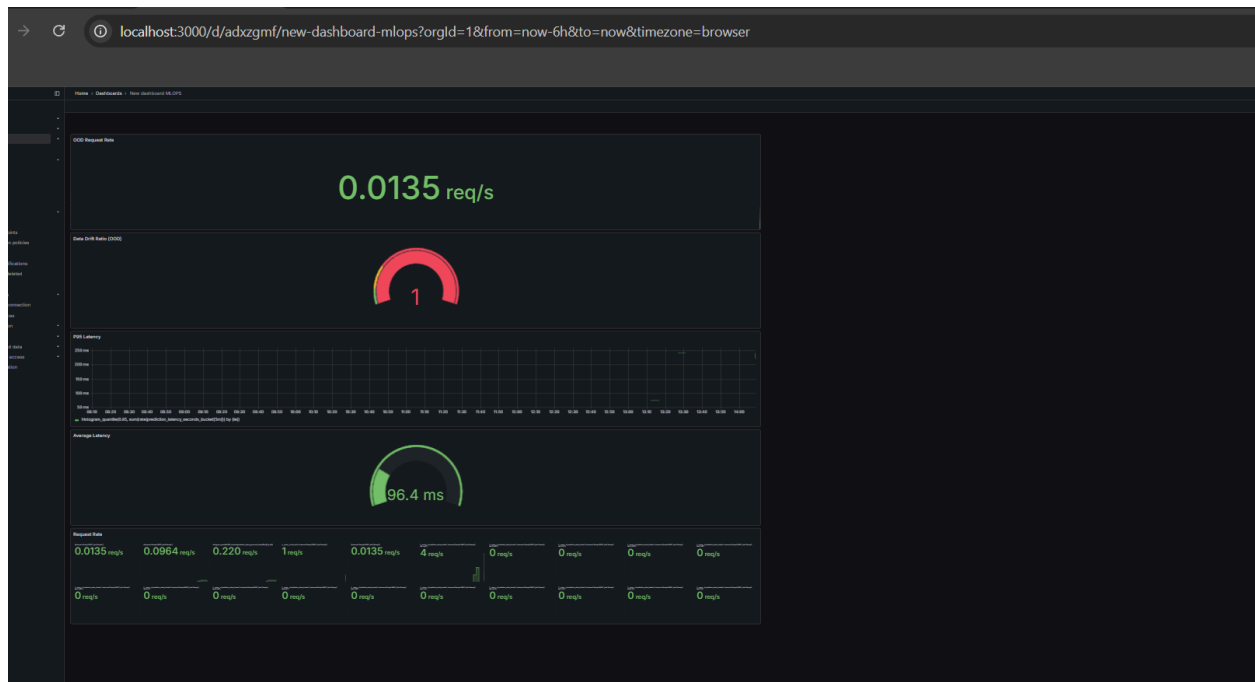
Status in prometheus



Graffana Dashboard



After requests

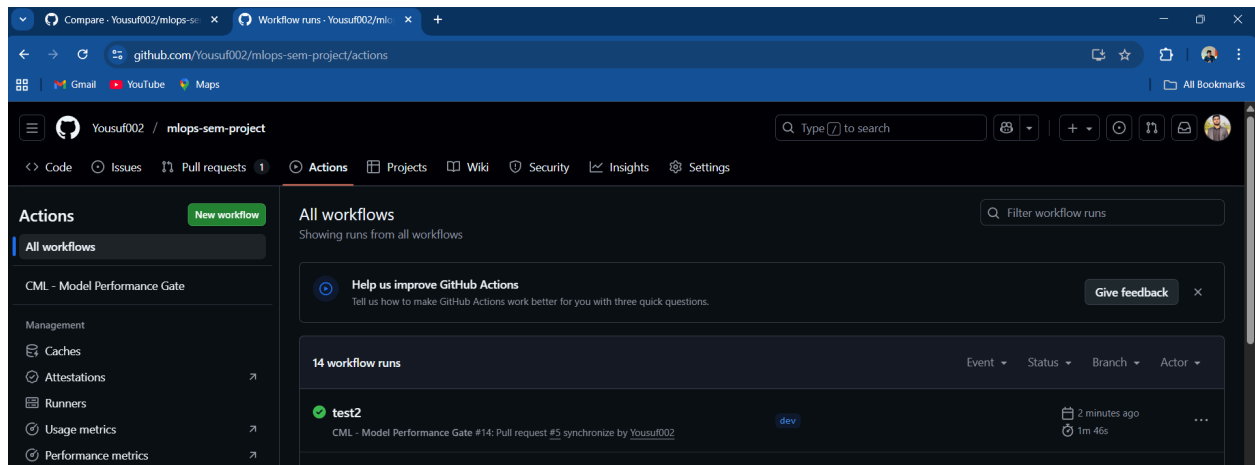


Creating pull request

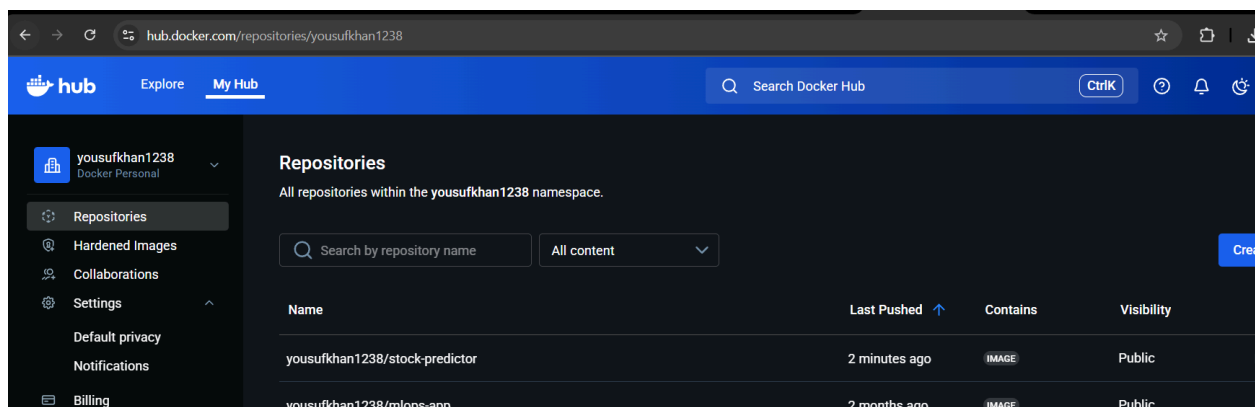
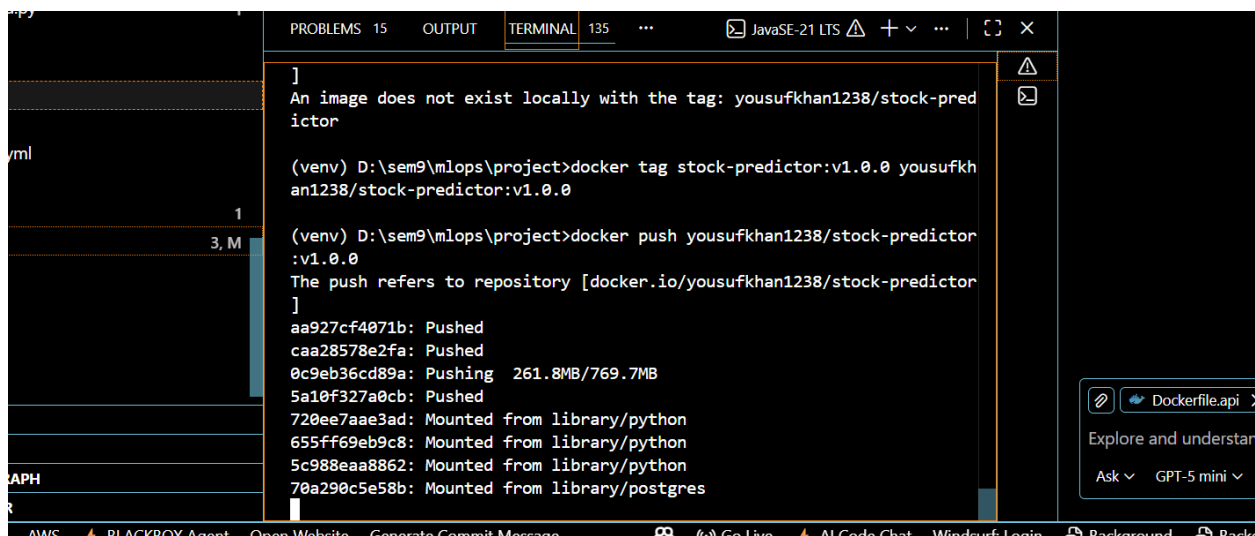
The screenshot shows a GitHub pull request page for the repository `mlops-sem-project`. The page displays the following information:

- Branches:** `base: test` and `compare: dev`. A green checkmark indicates "Able to merge. These branches can be automatically merged."
- Actions:** A green button labeled "Create pull request".
- Summary:** 3 commits, 35 files changed, 2 contributors.
- Commits:** A list of commits, including "Create cml.yml" by Yousuf002, "feat: complete monitoring + CML workflow" by yousufkhan02, and "Merge branch 'dev' of https://github.com/Yousuf002/mlops-sem-project" by yousufkhan02.
- Files:** A list of files changed, including `cml.yml` (24a75c9), `README.md` (f84120f), and `CONTRIBUTING.md` (eb8e402).

CML workflow



docker tag stock-predictor:v1.0.0 yousufkhan1238/stock-predictor:v1.0.0



Container running

